Do Now · retrieval

Lesson 113 · Open Source and Closed Source Decisions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Name a utility for recovering files after hardware loss.

Q2

What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q3

What is the time complexity of sequential search in the worst case?

Q4

After pushing A then B onto a stack, which is popped first?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Source availability — Open-source licences allow access to source code with specified reuse rights.

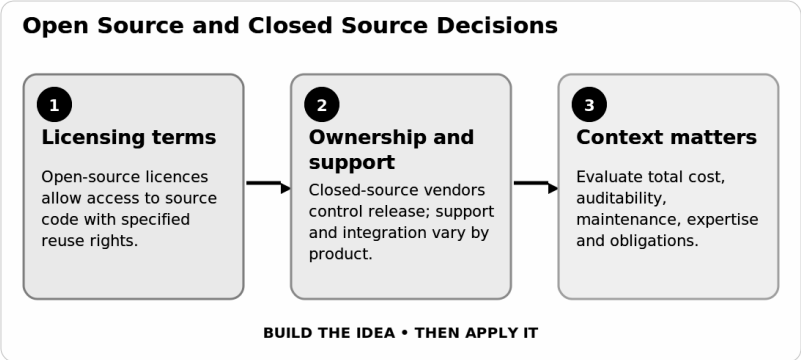
Core idea 2

Control vs support — Closed-source vendors control release; support and integration vary by product.

Core idea 3

Make a justified choice — Evaluate total cost, auditability, maintenance, expertise and obligations.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

open source — Code distributed under a licence allowing inspection and specified reuse.

closed source — Source and modification rights normally controlled by owner.

licence — Legal conditions attached to software use.

vendor lock-in — Difficult or expensive dependence on one supplier.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

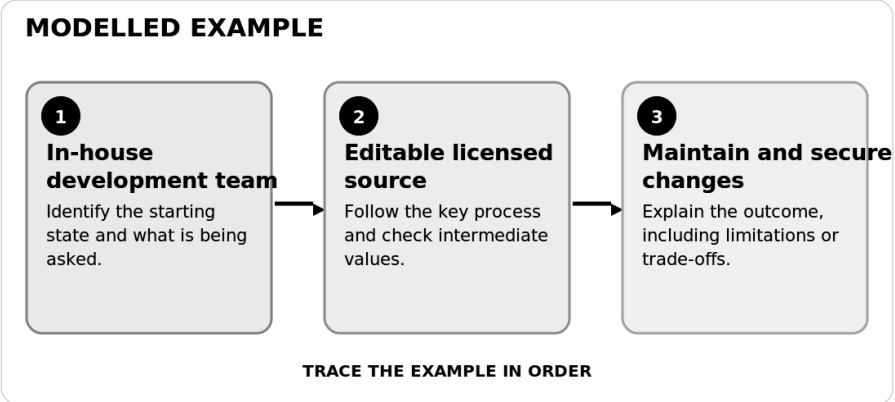
Problem

A council wants to adapt an accessible web service using its development team. Explain an open-source advantage and a risk.

Follow the worked method

1. Model answer — Relevant open-source licence can permit modification and inspection, allowing accessibility changes. Risk: the council must ensure support, security patching and licence compliance; source availability alone guarantees neither quality nor safety. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define open-source and closed-source software and state a licensing implication. [6]

Question 2

Explain why free-of-charge software is not necessarily open source. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate the choice for a charity's public website considering skills, privacy, costs and vendor dependence. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Open-source software is automatically more secure than closed-source software.

Use key vocabulary from memory

1. Explain open source in your own words.

2. Explain closed source in your own words.

3. Explain licence in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.